

Course code and name	ISYS2092/ISYS3397/EEET2578 Software Testing/Engineering Quality Assurance and Testing
Assignment name	Integrated Testing Strategy
Weight	40%
Type	Group (4-5 Students)
Presentation	[final teaching week - schedule to be confirmed]

1. Overview of Assessment

Working in a group of 4 to 5, you will complete the following:

1. Set up your group's personalised build of the system under test (SUT).
2. Produce a test plan, risk register, and consolidated defect summary (test management).
3. Stream A: web automation with Selenium WebDriver.
4. Stream B: performance testing with JMeter.
5. Stream C: BDD testing with Cucumber.
6. Report defects, write individual accountability sections and the AI usage and critique.
7. Submit the report and GitHub repository, and deliver a group presentation with individual Q&A.

2. Groups and the System Under Test

- Groups have 4 to 5 members. Register your group by [group registration deadline / method].
- **Build:** [download / GitHub Classroom link]. On setup, enter your group ID so the build is configured for your group.
- **Specification and non-functional requirements:** [specification file]. This is the single source of truth, and includes the performance thresholds used in Stream B.
- The build deviates from the specification in several places, different for each group, including at least one performance defect (for example an artificial delay or an inefficient query) that only appears under load.
- All screenshots and results must come from your group's own build.

3. Role Allocation

- Each member owns at least one stream (A, B, or C) and is the person who answers questions about it during the presentation.
- One member additionally takes the **Test Manager** role (Section 4).
- State your role allocation clearly at the front of the report.

4. Test Management Deliverable

- **Test plan:** scope, test approach for each stream, entry and exit criteria, and resource allocation across the team.
- **Risk register:** product and project risks, with likelihood, impact, and mitigation.
- **Consolidated defect summary:** all defects from the three streams, classified by severity and priority following ISTQB conventions.

5. Stream A: Web Automation (Selenium WebDriver, Java)

- Automate **at least 8 scenarios** against the SUT using Selenium WebDriver in Java.
- Each scenario must include **real assertions** that verify outcomes. Record-and-playback output alone is not acceptable.
- Organise the code sensibly (for example using the Page Object pattern) and push results and defects to Jira.

6. Stream B: Performance Testing (JMeter)

- Design load tests for **at least 3 key endpoints** of the SUT.
- Set the load profile (threads, ramp-up, duration) and define pass/fail thresholds based on the supplied non-functional requirements.
- Analyse the results, identify bottlenecks, and recommend improvements. Your analysis should locate the seeded performance defect that appears under load.

7. Stream C: BDD Testing (Cucumber)

- Write **at least 5 Gherkin scenarios** describing user behaviour, with working step definitions in IntelliJ.
- Ensure scenarios run and produce traceable pass/fail results, and link failures to defects.

8. Defect Reporting

Report each deviation found across the streams as a defect, each including: a clear summary and link to the detecting test, the violated requirement, expected versus actual behaviour, severity and priority with justification, and which stream detected it.

9. Contribution and Teamwork (read carefully)

9.1 Version control evidence

- All code (Selenium, JMeter test plans, Cucumber) lives in your group's GitHub Classroom repository: [link].
- Commit history is reviewed and must show sustained, meaningful contribution from each member over at least two weeks. A repository created or filled the night before the deadline is a red flag, and copy-paste commits without meaningful change do not count.

9.2 Peer assessment

- Each member confidentially rates every other member via **[TEAMMATES / Buddychек / the nominated tool]** on output quality, level of participation, and support for the team.
- Peer results adjust each member's mark by up to **±15% of the group mark**. Completing the peer assessment by the deadline is required.

9.3 Quality threshold

- Contribution that does not work or cannot be used does not count toward a member's contribution.
- A member whose usable contribution is clearly below **[agreed minimum, e.g. 20% of the group's effort]**, as evidenced by peer assessment and commit history, may have their individual mark capped at **[e.g. 50%]** of the group mark.

9.4 Individual accountability section

- Each member writes 200 to 300 words stating what they built, which ISTQB techniques they applied, and at least one significant testing decision they made.

10. AI Usage and Critique

- **Declaration:** list the AI tools the group used and what each was used for.
- **Critique:** identify at least 2 AI suggestions that were wrong, redundant, or invalid for your group's build, with execution evidence and the corrections made.

11. Presentation

- Format: **15 minutes of presentation plus 5 minutes of questions**.
- Demonstrate the testing on your group's own build (live or recorded), covering all three streams.
- **Each member presents their own stream** and answers questions about it. The Q&A is assessed individually, so be ready to explain your technical decisions.
- Reviewers may include the teaching team and an invited industry guest.

12. Report Structure

Your report must follow this exact section numbering.

1. Group and role allocation; build setup evidence (group ID).
2. Test plan and risk register.
3. Stream A: Selenium web automation.
4. Stream B: JMeter performance testing and analysis.
5. Stream C: Cucumber BDD.
6. Consolidated defect summary.
7. Individual accountability sections (one per member).
8. AI usage and critique.

Appendices (optional): full AI transcripts, extra evidence.

13. Submission Instructions

One submission per group on Canvas, unless stated otherwise:

1. GROUP-A3-Report.pdf: the group report following Section 12.
2. A link to the group's GitHub Classroom repository, on the first page of the report.
3. The presentation slides, and the demo video link if the demo is recorded.
4. Each member individually completes the peer assessment by [deadline].

Late submissions are deducted 10% per day. After 5 days, zero will be granted, unless special consideration has been granted.

14. Marking

The group deliverable is assessed against the rubric in Appendix B (total 40 points). The resulting group mark is then adjusted for each member by the peer-assessment factor (up to $\pm 15\%$) and, where applicable, the quality-threshold cap described in Section 9.

Appendix A: Role and Contribution Summary Template

Complete one row per member. This sits alongside the individual accountability sections in the report.

Member (name, student ID)	Stream / role	Main deliverables produced	Key testing decision made	Commits / evidence ref